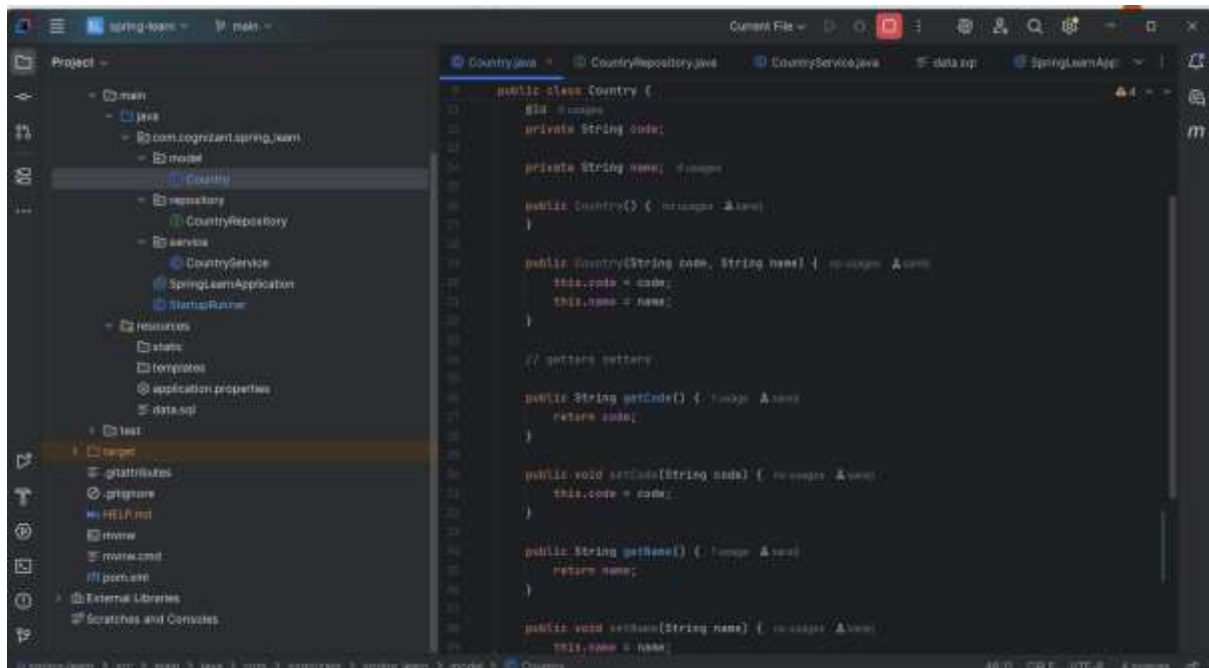
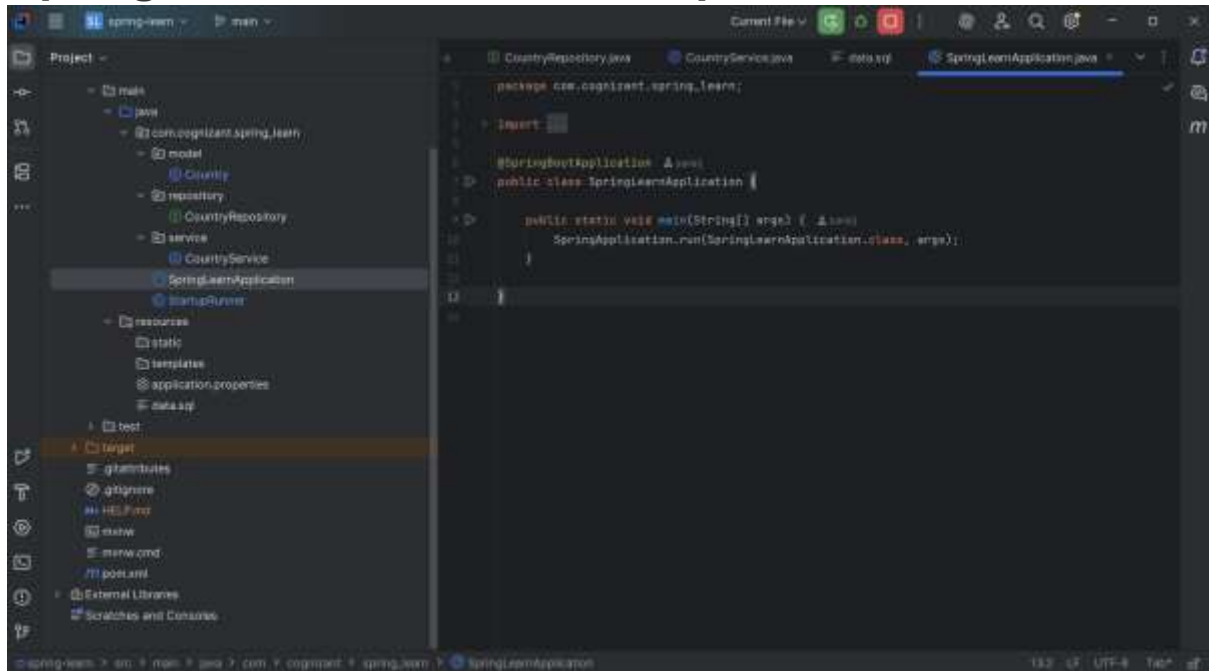
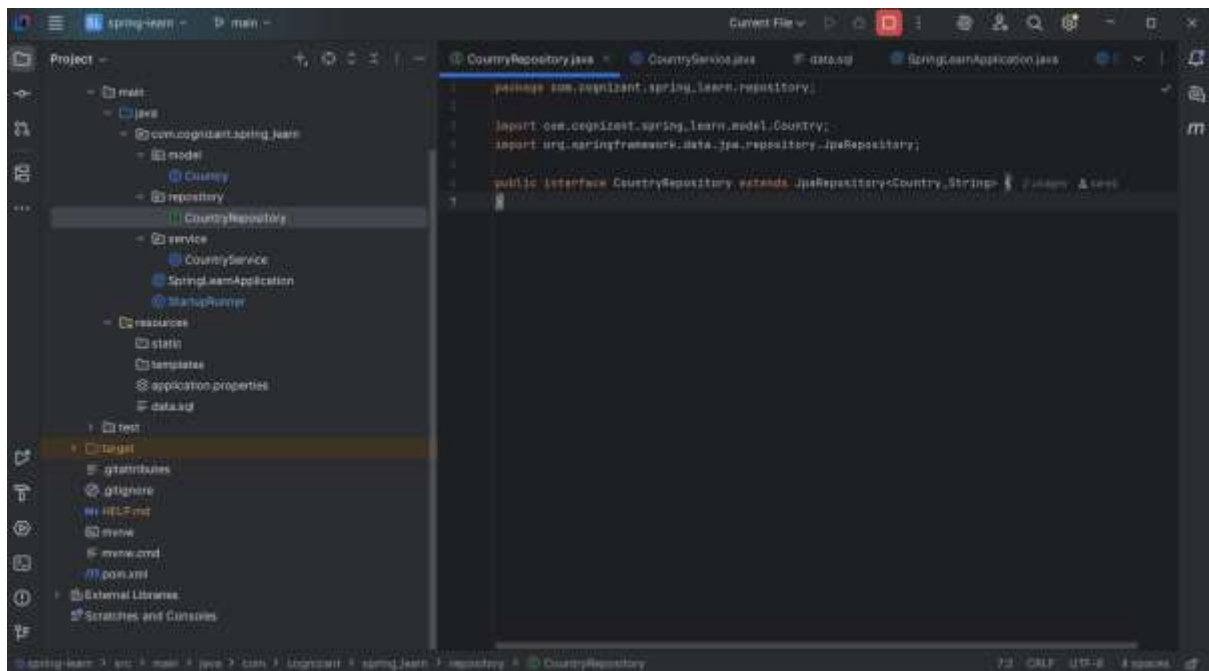
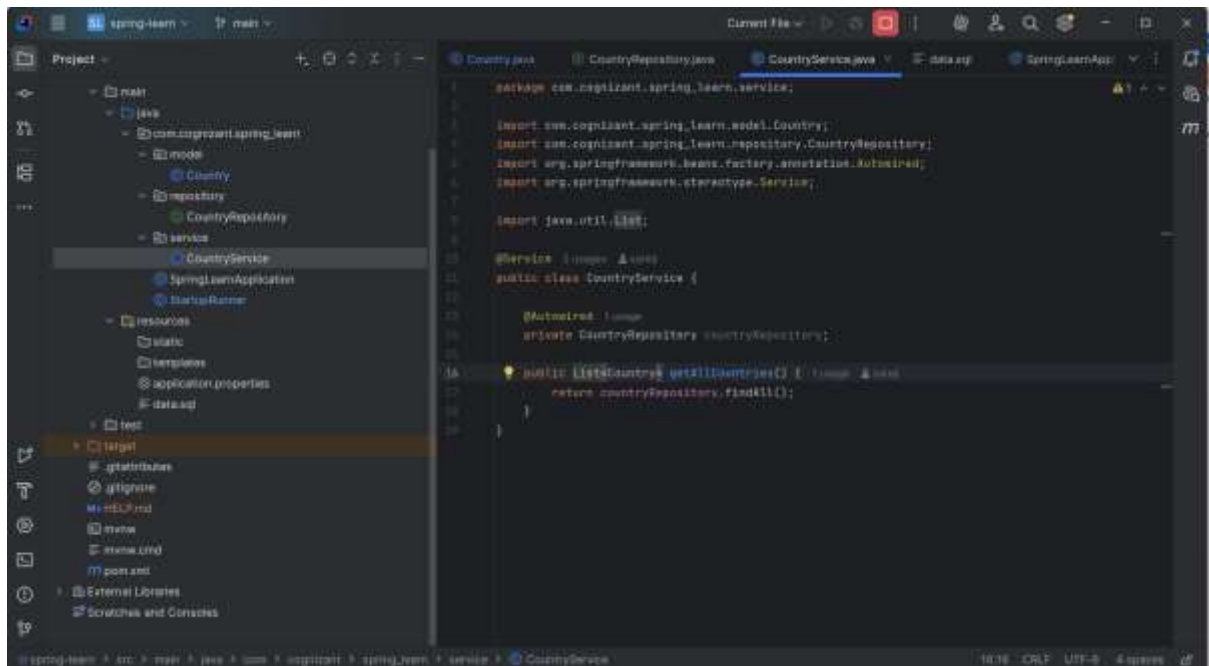
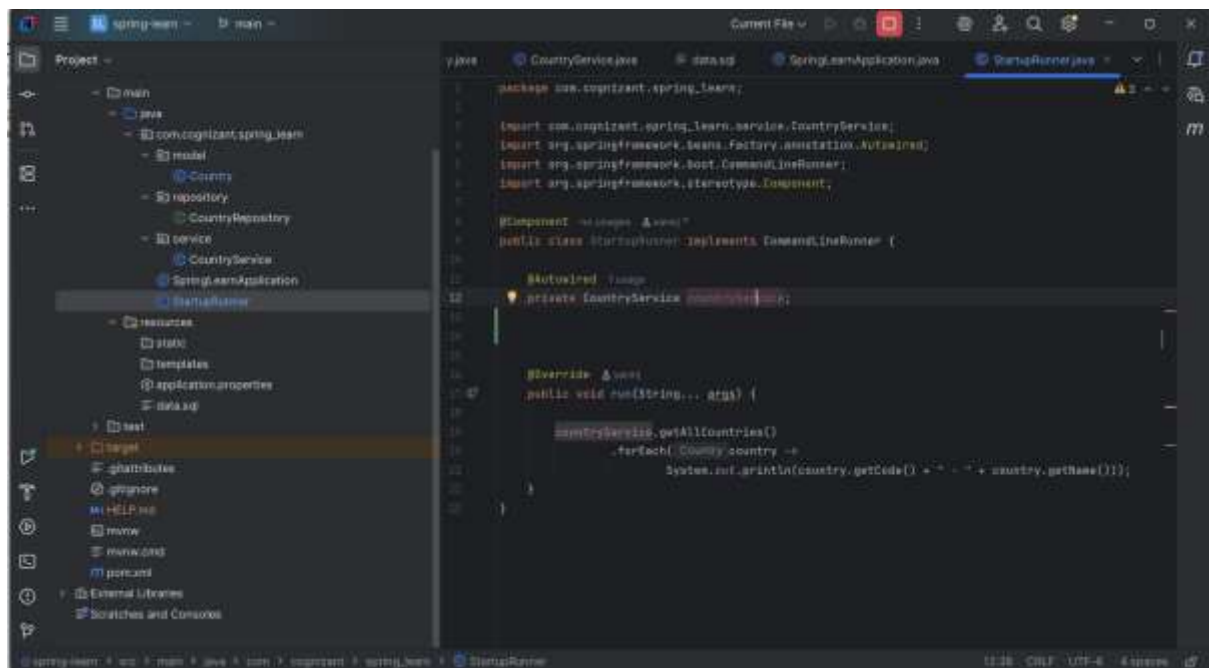
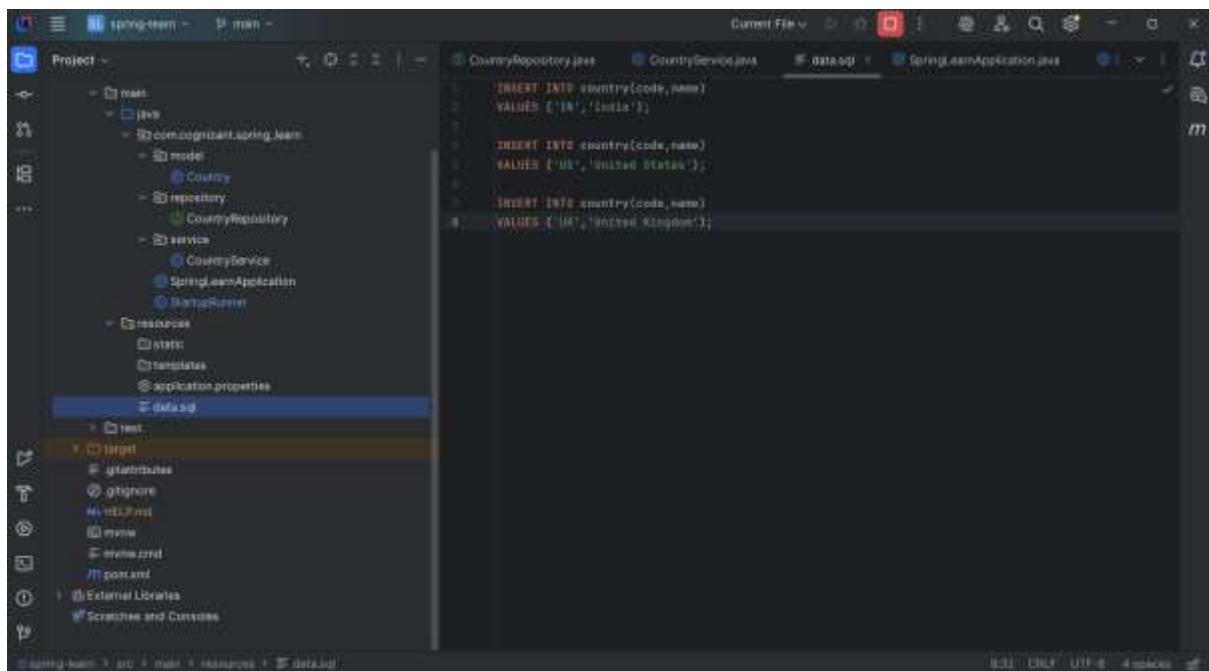


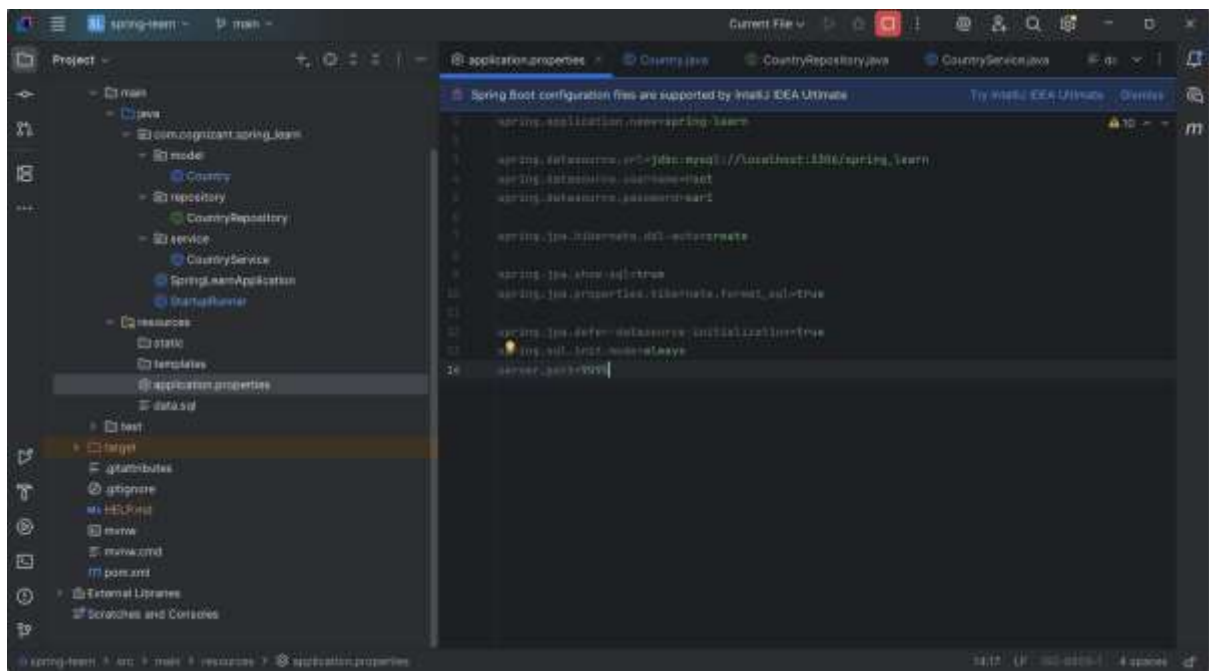
Hands on 1

Spring Data JPA - Quick Example

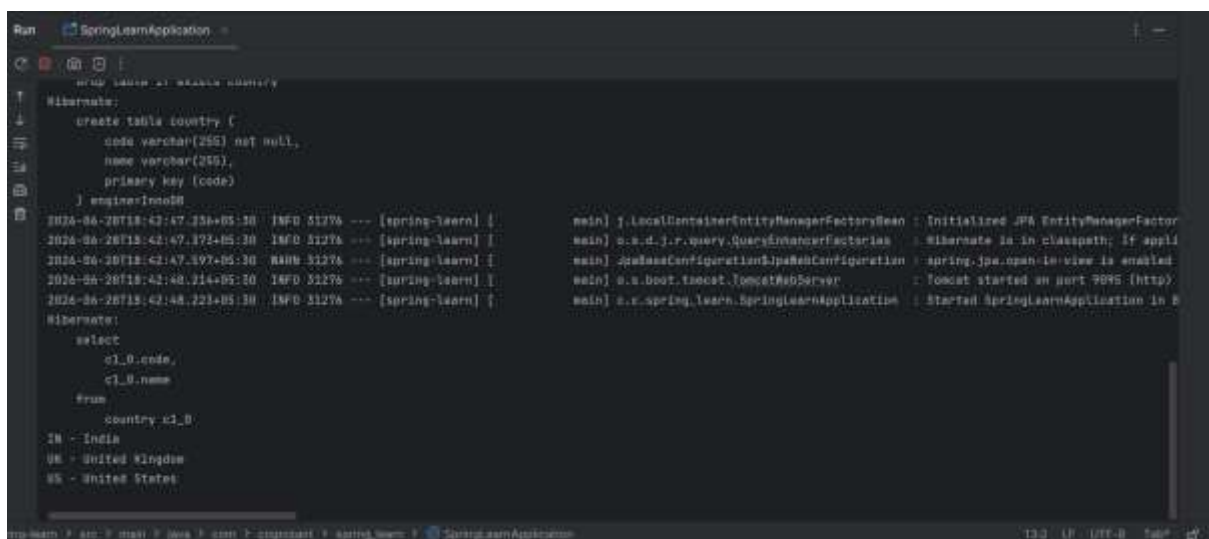








OUTPUT:



Handson:5 Implement services for managing Country

```
import ...

@SpringBootApplication
public class OrmLearnApplication {

    private static CountryService countryService; // usages

    public static void main(String[] args) {

        ApplicationContext context =
            SpringApplication.run(OrmLearnApplication.class, args);

        countryService = context.getBean(CountryService.class);
        // testAddCountry();
        // testUpdateCountry();
        // testDeleteCountry();
        // testFindCountriesByPartialName();
        // testGetCountry();
    }

    private static void testFindCountriesByName() { // no usages

        List<Country> countries =
            countryService.findCountriesByPartialName(text "ou");

        countries.forEach(System.out::println);
    }

    private static void testFindCountriesByPartialName() { // no usages
```

```

public class OrmLearnApplication {
    }
    private static void testSearchCountry() { no usages

        List<Country> countries =
            countryService.searchCountries(text: "un");

        System.out.println("Size = " + countries.size());

        countries.forEach(System.out::println);
    }
    // private static void testSearchCountry() {
    //
    //     List<Country> countries =
    //         countryService.searchCountries("an");
    //
    //     countries.forEach(System.out::println);
    // }
    private static void testAddCountry() { no usages

        Country country = new Country();

        country.setCode("NP");
        country.setName("Nepal");

        countryService.addCountry(country);

        System.out.println("Country Added");
    }
    private static void testGetCountry() { no usages

```

```

public class OrmLearnApplication {
    private static void testGetCountry() { no usages

        System.out.println("Start");

        Country country = countryService.findCountryByCode("IN");

        System.out.println(country);

        System.out.println("End");
    }
    private static void testUpdateCountry() { no usages

        Country country = countryService.findCountryByCode("IN");

        country.setName("India Updated");

        countryService.updateCountry(country);

        System.out.println("Country Updated");
    }
    private static void testDeleteCountry() { no usages

        countryService.deleteCountry(code: "AX");

        System.out.println("Country Deleted");
    }
}

```

```

package com.cognizant.orm_learn.model;

import ...

@Entity
@Table(name = "country")
public class Country {

    @Id
    @Column(name = "co_code")
    private String code;

    @Column(name = "co_name")
    private String name;

    public Country() {
    }

    public String getCode() { return code; }

    public void setCode(String code) { this.code = code; }

    public String getName() { return name; }

    public void setName(String name) { this.name = name; }

    @Override
    public String toString() { return "Country [code=" + code + ", name=" + name + "];" }
}

```

```

package com.cognizant.orm_learn.repository;

import ...

@Repository
public interface CountryRepository extends JpaRepository<Country, String> {

    Country findByCode(String code);

    List<Country> findByNameContainingIgnoreCase(String name);
}

```



```

@Service
public class CountryService {

    @Autowired
    private CountryRepository countryRepository;

    @Transactional
    public List<Country> getAllCountries() { return countryRepository.findAll(); }

    @Transactional
    public Country findCountryByCode(String code) { return countryRepository.findByCode(code); }

    @Transactional
    public Country addCountry(Country country) { return countryRepository.save(country); }

    @Transactional
    public Country updateCountry(Country country) { return countryRepository.save(country); }

    @Transactional
    public void deleteCountry(String code) { countryRepository.deleteByCode(code); }

    @Transactional
    public List<Country> searchCountries(String text) { return countryRepository.findByNameContainingIgnoreCase(text); }

    @Transactional
    public List<Country> findCountriesByPartialName(String text) {
        return countryRepository.findByNameContainingIgnoreCase(text);
    }
}

```

OUTPUT:

Get All Countries

```

24-06-26 18:55:24.403 restartedMain INFO org.hibernate.org.core InitializeService 55 HH0000489: No JPA platform available (set 'hibernat
24-06-26 18:55:24.537 restartedMain INFO org.hibernate.org.core buildNativeEntityManagerFactory 464 Initialized JPA EntityManagerFactory for pers
24-06-26 18:55:24.721 restartedMain INFO org.hibernate.org.core <init> 60 Hibernate is in classpath; if applicable, SQL parse
24-06-26 18:55:25.129 restartedMain WARN ation$JpaWebConfiguration openEntityManagerInViewInterceptor 261 spring.jpa.open-in-view is enabled by defe
24-06-26 18:55:26.154 restartedMain INFO o.s.b.t.TomcatWebServer start 240 Tomcat started on port 9095 (http) with context pat
24-06-26 18:55:26.178 restartedMain INFO o.s.s.OrelearnApplication logStarted 60 Started OrelearnApplication in 12.22 seconds (proce
Start
24-06-26 18:55:26.187 restartedMain DEBUG org.hibernate.SQL logStatement 126 select c1_0.code,c1_0.name from country c1_0 where
Country {code=IN, name=India}
End

```

Find Country By Code

```

24-06-26 11:05:04.139 restartedMain INFO org.hibernate.org.core <init> 60 Hibernate is in classpath; if applicable, SQL parse
24-06-26 11:05:04.139 restartedMain WARN ation$JpaWebConfiguration openEntityManagerInViewInterceptor 261 spring.jpa.open-in-view is enabled by defe
24-06-26 11:05:05.066 restartedMain INFO o.s.b.t.TomcatWebServer start 240 Tomcat started on port 9095 (http) with context pat
24-06-26 11:05:05.088 restartedMain INFO o.s.s.OrelearnApplication logStarted 60 Started OrelearnApplication in 11.235 seconds (pro
Start
24-06-26 11:05:06.063 restartedMain DEBUG org.hibernate.SQL logStatement 126 select c1_0.code,c1_0.name from country c1_0 where
Country {code=IN, name=India}
End

```

Add Country


```
24-06-26 11:13:52.900 restartedMain INFO org.hibernate.engine.jdbc.connections.internal.DriverManagerConnectionProviderImpl new initialized JPA EntityManagerFactory for pers
24-06-26 11:13:53.377 restartedMain INFO .q.QueryEnhancerFactoryes <clinit> 48 Hibernate is in classpath; If applicable, HQL parse
WARN ation$JpaWebConfiguration openEntityManagerInViewInterceptor 261 spring.jpa.open-in-view is enabled by defa
24-06-26 11:13:54.341 restartedMain INFO o.s.h.t.TomcatWebServer start 240 Tomcat started on port 9095 (http) with context pa
24-06-26 11:13:54.357 restartedMain INFO v.c.c.GrmLearnApplication logStarted 48 Started GrmLearnApplication in 11.882 seconds (proc
24-06-26 11:13:54.484 restartedMain DEBUG org.hibernate.SQL logStatement 126 select c1_0.code,c1_0.name from country c1_0 where
24-06-26 11:13:54.485 restartedMain DEBUG org.hibernate.SQL logStatement 126 insert into country (name,code) values (?,?)
Country Added
```

UpdateCountry

```
Run GrmLearnApplication
24-06-26 17:21:16.004 restartedMain INFO .q.QueryEnhancerFactoryes <clinit> 48 Hibernate is in classpath; If applicable, HQL parse
24-06-26 17:21:16.385 restartedMain WARN ation$JpaWebConfiguration openEntityManagerInViewInterceptor 261 spring.jpa.open-in-view is enabled by defa
24-06-26 17:21:17.937 restartedMain INFO o.s.h.t.TomcatWebServer start 240 Tomcat started on port 9095 (http) with context pa
24-06-26 17:21:17.957 restartedMain INFO v.c.c.GrmLearnApplication logStarted 48 Started GrmLearnApplication in 18.095 seconds (proc
24-06-26 17:21:17.863 restartedMain DEBUG org.hibernate.SQL logStatement 126 select c1_0.code,c1_0.name from country c1_0 where
24-06-26 17:21:17.941 restartedMain DEBUG org.hibernate.SQL logStatement 126 select c1_0.code,c1_0.name from country c1_0 where
24-06-26 17:21:17.973 restartedMain DEBUG org.hibernate.SQL logStatement 126 update country set name=? where code=?
Country Updated
```

DeleteCountry

```
Run GrmLearnApplication
24-06-26 17:29:05.414 restartedMain INFO org.hibernate.engine.jdbc.connections.internal.DriverManagerConnectionProviderImpl new initialized JPA EntityManagerFactory for pers
24-06-26 17:29:05.438 restartedMain INFO rEntityManagerFactory$ean buildNativeEntityManagerFactory 464 Initialized JPA EntityManagerFactory for pers
24-06-26 17:29:05.657 restartedMain INFO .q.QueryEnhancerFactoryes <clinit> 48 Hibernate is in classpath; If applicable, HQL parse
24-06-26 17:29:06.418 restartedMain WARN ation$JpaWebConfiguration openEntityManagerInViewInterceptor 261 spring.jpa.open-in-view is enabled by defa
24-06-26 17:29:07.111 restartedMain INFO o.s.h.t.TomcatWebServer start 240 Tomcat started on port 9095 (http) with context pa
24-06-26 17:29:07.929 restartedMain INFO v.c.c.GrmLearnApplication logStarted 48 Started GrmLearnApplication in 13.681 seconds (proc
24-06-26 17:29:07.637 restartedMain DEBUG org.hibernate.SQL logStatement 126 select c1_0.code,c1_0.name from country c1_0 where
Country Deleted
```

Find list of countries matching a partial country name

```
24-06-26 18:34:46.388 restartedMain INFO org.hibernate.engine.jdbc.connections.internal.DriverManagerConnectionProviderImpl new initialized JPA EntityManagerFactory for pers
24-06-26 18:34:46.351 restartedMain INFO rEntityManagerFactory$ean buildNativeEntityManagerFactory 464 Initialized JPA EntityManagerFactory for pers
24-06-26 18:34:46.470 restartedMain INFO .q.QueryEnhancerFactoryes <clinit> 48 Hibernate is in classpath; If applicable, HQL parse
24-06-26 18:34:46.864 restartedMain WARN ation$JpaWebConfiguration openEntityManagerInViewInterceptor 261 spring.jpa.open-in-view is enabled by defa
24-06-26 18:34:47.412 restartedMain INFO o.s.h.t.TomcatWebServer start 240 Tomcat started on port 9095 (http) with context pa
24-06-26 18:34:47.422 restartedMain INFO v.c.c.GrmLearnApplication logStarted 48 Started GrmLearnApplication in 7.72 seconds (proc
24-06-26 18:34:48.060 restartedMain DEBUG org.hibernate.SQL logStatement 126 select c1_0.co_code,c1_0.co_name from country c1_0
Country [code=BV, name=Bouvet Island]
Country [code=GP, name=Guadeloupe]
Country [code=LU, name=Luxembourg]
Country [code=SS, name=South Sudan]
Country [code=TF, name=French Southern Territories]
Country [code=UM, name=United States Minor Outlying Islands]
Country [code=ZA, name=South Africa]
```

Hands on 6

Find a country based on country code

```
@SpringBootApplication
public class OrmLearnApplication {

    private static CountryService countryService;
    public static void main(String[] args)
        throws CountryNotFoundException {

        ApplicationContext context =
            SpringApplication.run(
                OrmLearnApplication.class, args);

        countryService =
            context.getBean(CountryService.class);

        getAllCountriesTest();
    }
    private static void getAllCountriesTest()
        throws CountryNotFoundException {

        System.out.println("Start");

        Country country =
            countryService.findCountryByCode( countryCode: "IN");

        System.out.println(country);

        System.out.println("End");
    }
}
```

```
spring.application.name=orm-learn
# Spring Framework and application log
logging.level.org.springframework=info
logging.level.com.cognizant=debug

# Hibernate logs
logging.level.org.hibernate.SQL=trace
logging.level.org.hibernate.type.descriptor.sql=trace

# Log pattern
logging.pattern.console=%d{dd-MM-yy} %d{HH:mm:ss.SSS} %-20.20thread %5p %-25.25logger{25} %25M %4

# Database configuration
spring.datasource.driver-class-name=com.mysql.cj.jdbc.Driver
spring.datasource.url=jdbc:mysql://localhost:3306/ormlearn
spring.datasource.username=root
spring.datasource.password=sarl

# Hibernate configuration
spring.jpa.hibernate.ddl-auto=validate
spring.jpa.properties.hibernate.dialect=org.hibernate.dialect.MySQLDialect

server.port=9095
```

```

3 public class CountryService {
4     public Country findCountryByCode(String countryCode)
5     {
6         Country country = result.get();
7
8         return country;
9     }
10
11     @Transactional no usages
12     > public Country addCountry(Country country) { return countryRepository.save(country); }
13
14     @Transactional no usages
15     > public Country updateCountry(Country country) { return countryRepository.save(country); }
16
17     @Transactional no usages
18     > public void deleteCountry(String code) { countryRepository.deleteById(code); }
19
20     @Transactional no usages
21     > public List<Country> searchCountries(String text) { return countryRepository.findByNameContaining(text); }
22
23     @Transactional no usages
24     public List<Country> findCountriesByPartialName(String text) {
25         return countryRepository.findByNameContainingIgnoreCase(text);
26     }
27 }

```

```

1 package com.cognizant.orm_learn.repository;
2
3 import com.cognizant.orm_learn.model.Country;
4 import org.springframework.data.jpa.repository.JpaRepository;
5 import org.springframework.stereotype.Repository;
6
7 import java.util.List;
8
9 @Repository 2 usages
10 public interface CountryRepository extends JpaRepository<Country, String> {
11
12     Country findByCode(String code); no usages
13
14     List<Country> findByNameContainingIgnoreCase(String name); 2 usages
15
16 }
17
18

```

```
pool: pool size: 1000
Pool: DataSourceConnectionProvider
Minimum pool size: undefined/unknown
Maximum pool size: undefined/unknown
24-06-26 18:52:04.560 restartedMain INFO org.hibernate.orm.core initiateService 33 HH0000489: No JTA platform available (set "hibernat
24-06-26 18:52:04.637 restartedMain INFO org.hibernate.orm.core buildNativeEntityManagerFactory 464 Initialized JPA EntityManagerFactory for pers
24-06-26 18:52:04.775 restartedMain INFO org.hibernate.orm.core <clinit> 48 Hibernate is in classpath; if applicable, HQL parse
24-06-26 18:52:05.163 restartedMain INFO org.hibernate.orm.core openEntityManagerInViewInterceptor 261 Spring Jpa.open-in-view is enabled by defa
24-06-26 18:52:05.764 restartedMain INFO org.hibernate.orm.core start 340 JUnit started on port 3091 (http) with context pat
24-06-26 18:52:05.764 restartedMain INFO org.hibernate.orm.core logStarted 40 Started GrailsApplication in 10.955 seconds (proc
Start
24-06-26 18:52:05.822 restartedMain INFO org.hibernate.orm.core logStatement 124 select ci_0.co_code,ci_0.co_name from country ci_0
Country (codeID, nameIndia)
End
```

DATABASE SCREENSHOTS

```
1 * USE ormlearn;
2 * DELETE FROM country;
3
4 * CREATE TABLE country (
5     co_code VARCHAR(2) PRIMARY KEY,
6     co_name VARCHAR(100) NOT NULL
7 );
8
9 * insert into country (co_code, co_name) values ("AF", "Afghanistan");
10 * insert into country (co_code, co_name) values ("AL", "Albania");
11 * insert into country (co_code, co_name) values ("DZ", "Algeria");
12 * insert into country (co_code, co_name) values ("AS", "American Samoa");
13 * insert into country (co_code, co_name) values ("AD", "Andorra");
14 * insert into country (co_code, co_name) values ("AO", "Angola");
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39 *
40 * insert into country (co_code, co_name) values ("IO", "British Indian Ocean Territory");
41 * insert into country (co_code, co_name) values ("BN", "Brunei Darussalam");
42 * insert into country (co_code, co_name) values ("BG", "Bulgaria");
43 * insert into country (co_code, co_name) values ("BF", "Burkina Faso");
44 * insert into country (co_code, co_name) values ("BI", "Burundi");
45 * insert into country (co_code, co_name) values ("KH", "Cambodia");
46 * insert into country (co_code, co_name) values ("CM", "Cameroon");
47 * insert into country (co_code, co_name) values ("CA", "Canada");
48 * insert into country (co_code, co_name) values ("CV", "Cape Verde");
49 * insert into country (co_code, co_name) values ("KY", "Cayman Islands");
50 * insert into country (co_code, co_name) values ("CF", "Central African Republic");
51 * insert into country (co_code, co_name) values ("TD", "Chad");
52 * insert into country (co_code, co_name) values ("CL", "Chile");
53 * insert into country (co_code, co_name) values ("CN", "China");
54 *
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```
Limit to 1000 rows
67 * insert into country (co_code, co_name) values ("CI", "Côte d'Ivoire");
68 * insert into country (co_code, co_name) values ("DK", "Denmark");
69 * insert into country (co_code, co_name) values ("DJ", "Djibouti");
70 * insert into country (co_code, co_name) values ("DM", "Dominica");
71 * insert into country (co_code, co_name) values ("DO", "Dominican Republic");
72 * insert into country (co_code, co_name) values ("EC", "Ecuador");
73 * insert into country (co_code, co_name) values ("EG", "Egypt");
74 * insert into country (co_code, co_name) values ("SV", "El Salvador");
75 * insert into country (co_code, co_name) values ("GQ", "Equatorial Guinea");
76 * insert into country (co_code, co_name) values ("ER", "Eritrea");
77 * insert into country (co_code, co_name) values ("EE", "Estonia");
78 * insert into country (co_code, co_name) values ("ET", "Ethiopia");
79 * insert into country (co_code, co_name) values ("FK", "Falkland Islands (Malvinas)");
80 * insert into country (co_code, co_name) values ("FO", "Faroe Islands");
```

```
79 * insert into country (co_code, co_name) values ("FK", "Falkland Islands (Malvinas)");
80 * insert into country (co_code, co_name) values ("FO", "Faroe Islands");
81 * insert into country (co_code, co_name) values ("FJ", "Fiji");
82 * insert into country (co_code, co_name) values ("FI", "Finland");
83 * insert into country (co_code, co_name) values ("FR", "France");
84 * insert into country (co_code, co_name) values ("GF", "French Guiana");
85 * insert into country (co_code, co_name) values ("PF", "French Polynesia");
86 * insert into country (co_code, co_name) values ("TF", "French Southern Territories");
87 * insert into country (co_code, co_name) values ("GA", "Gabon");
88 * insert into country (co_code, co_name) values ("GM", "Gambia");
89 * insert into country (co_code, co_name) values ("GE", "Georgia");
90 * insert into country (co_code, co_name) values ("DE", "Germany");
91 * insert into country (co_code, co_name) values ("GH", "Ghana");
92 * insert into country (co_code, co_name) values ("GI", "Gibraltar");
```

```
100 * insert into country (co_code, co_name) values ("GN", "Guinea");
101 * insert into country (co_code, co_name) values ("GW", "Guinea-Bissau");
102 * insert into country (co_code, co_name) values ("GY", "Guyana");
103 * insert into country (co_code, co_name) values ("HT", "Haiti");
104 * insert into country (co_code, co_name) values ("HM", "Heard Island and McDonald Islands");
105 * insert into country (co_code, co_name) values ("VA", "Holy See (Vatican City State)");
106 * insert into country (co_code, co_name) values ("HN", "Honduras");
107 * insert into country (co_code, co_name) values ("HK", "Hong Kong");
108 * insert into country (co_code, co_name) values ("HU", "Hungary");
109 * insert into country (co_code, co_name) values ("IS", "Iceland");
110 * insert into country (co_code, co_name) values ("IN", "India");
111 * insert into country (co_code, co_name) values ("ID", "Indonesia");
112 * insert into country (co_code, co_name) values ("IR", "Iran, Islamic Republic of");
113 * insert into country (co_code, co_name) values ("IQ", "Iraq");
```

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Window Help

Navigator

SCHEMAS

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Schema: testdb

SQL File 7 SQL File 8 SQL File 9 SQL File 10 SQL File 11 SQL File 12 SQL File 13 SQL File 14 SQL File 15 SQL File 16 SQL File 17 SQL File 18

137 * Insert into country (co_code, co_name) values ('17', 'Czechia');
138 * Insert into country (co_code, co_name) values ('18', 'Czechia');
139 * Insert into country (co_code, co_name) values ('19', 'Czechia');
140 * Insert into country (co_code, co_name) values ('20', 'Czechia');
141 * Insert into country (co_code, co_name) values ('21', 'Czechia');
142 * Insert into country (co_code, co_name) values ('22', 'Czechia');
143 * Insert into country (co_code, co_name) values ('23', 'Czechia');
144 * Insert into country (co_code, co_name) values ('24', 'Czechia');
145 * Insert into country (co_code, co_name) values ('25', 'Czechia');
146 * Insert into country (co_code, co_name) values ('26', 'Czechia');
147 * Insert into country (co_code, co_name) values ('27', 'Czechia');
148 * Insert into country (co_code, co_name) values ('28', 'Czechia');
149 * Insert into country (co_code, co_name) values ('29', 'Czechia');
150 * Insert into country (co_code, co_name) values ('30', 'Czechia');
151 * Insert into country (co_code, co_name) values ('31', 'Czechia');
152 * Insert into country (co_code, co_name) values ('32', 'Czechia');
153 * Insert into country (co_code, co_name) values ('33', 'Czechia');
154 * Insert into country (co_code, co_name) values ('34', 'Czechia');
155 * Insert into country (co_code, co_name) values ('35', 'Czechia');
156 * Insert into country (co_code, co_name) values ('36', 'Czechia');
157 * Insert into country (co_code, co_name) values ('37', 'Czechia');
158 * Insert into country (co_code, co_name) values ('38', 'Czechia');
159 * Insert into country (co_code, co_name) values ('39', 'Czechia');
160 * Insert into country (co_code, co_name) values ('40', 'Czechia');
161 * Insert into country (co_code, co_name) values ('41', 'Czechia');
162 * Insert into country (co_code, co_name) values ('42', 'Czechia');
163 * Insert into country (co_code, co_name) values ('43', 'Czechia');
164 * Insert into country (co_code, co_name) values ('44', 'Czechia');
165 * Insert into country (co_code, co_name) values ('45', 'Czechia');
166 * Insert into country (co_code, co_name) values ('46', 'Czechia');
167 * Insert into country (co_code, co_name) values ('47', 'Czechia');
168 * Insert into country (co_code, co_name) values ('48', 'Czechia');
169 * Insert into country (co_code, co_name) values ('49', 'Czechia');
170 * Insert into country (co_code, co_name) values ('50', 'Czechia');

Result Grid

Country

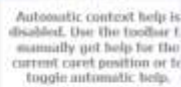
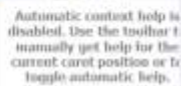
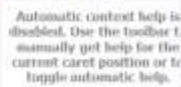
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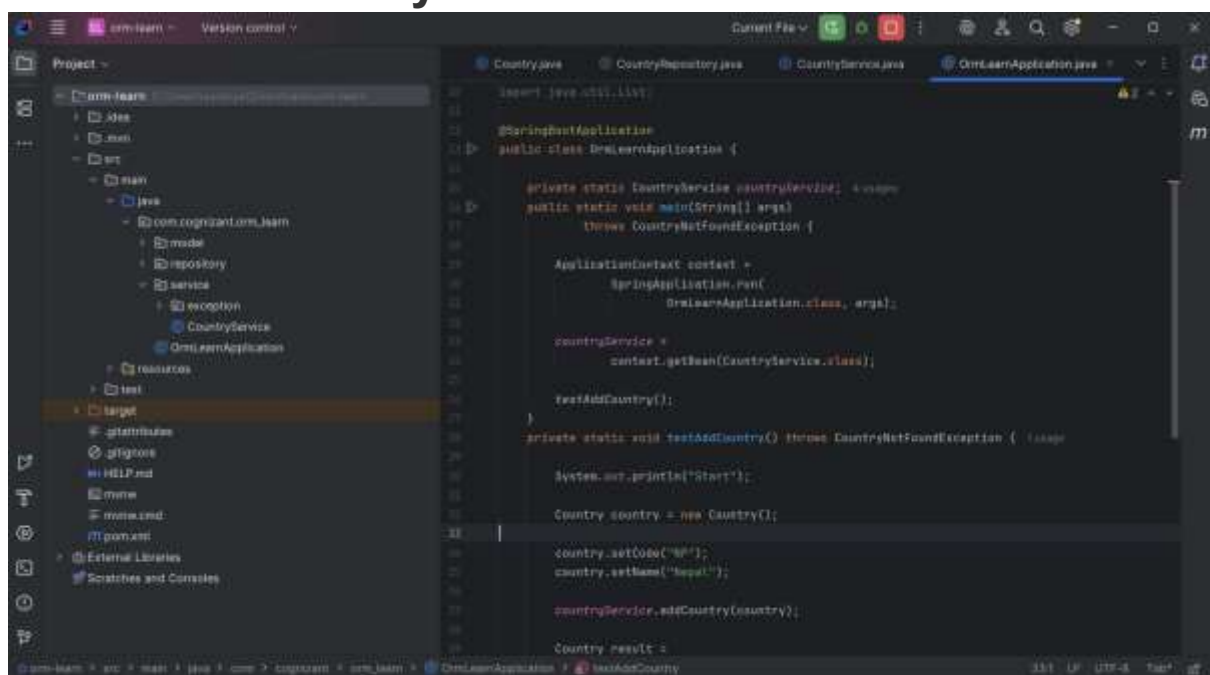
Result 3

Active Output

Time	Query	Rows	Storage	Duration / Rows
132 10:21:22	SELECT COUNT(*) FROM country LIMIT 1, 1000	1 row returned		0.019 sec / 0.000 sec
133 10:16:52	SELECT * FROM country WHERE co_code=100 LIMIT 1, 1000	1 row returned		0.000 sec / 0.000 sec
134 10:21:00	SELECT COUNT(*) FROM country LIMIT 1, 1000	1 row returned		0.019 sec / 0.000 sec

Automatic content help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.





```
Country.java CountryRepository.java CountryService.java OmlLearnApplication.java
public class CountryService {
    @Autowired
    private CountryRepository countryRepository;

    @Transactional
    public List<Country> getCountryList() {
        return countryRepository.findAll();
    }

    @Transactional
    public Country findCountryByCode(String countryCode)
        throws CountryNotFoundException {
        Optional<Country> result =
            countryRepository.findById(countryCode);

        if (!result.isPresent()) {
            throw new CountryNotFoundException(
                "Country not found");
        }

        Country country = result.get();

        return country;
    }

    @Transactional
    public void saveCountry(Country country) {
        countryRepository.save(country);
    }
}
```

```
Country.java CountryRepository.java CountryService.java OmlLearnApplication.java
public class CountryService {
    public void saveCountry(Country country) {
        countryRepository.save(country);
    }

    @Transactional
    public Country updateCountry(Country country) {
        return countryRepository.save(country);
    }

    @Transactional
    public void deleteCountry(String code) {
        countryRepository.deleteById(code);
    }

    @Transactional
    public List<Country> searchCountryList(String text) {
        return countryRepository.findAllContainingIgnoreCase(text);
    }

    @Transactional
    public List<Country> findCountryListByPart(String text) {
        return countryRepository.findAllContainingIgnoreCase(text);
    }
}
```

```
1 package com.cognizant.orn.learn.repository;
2
3 import com.cognizant.orn.learn.model.Country;
4 import org.springframework.data.jpa.repository.JpaRepository;
5 import org.springframework.stereotype.Repository;
6
7 import java.util.List;
8
9 @Repository
10 public interface CountryRepository extends JpaRepository<Country, String> {
11
12     Country findCountryByCode(String code);
13
14     List<Country> findByNameContainingIgnoreCase(String name);
15
16 }
```

```
1 @Entity
2 @Table(name = "country")
3 public class Country {
4
5     @Id
6     @GeneratedValue(strategy = GenerationType.IDENTITY)
7     private String code;
8
9     private String name;
10
11     public Country() {}
12
13     public String getCountry() {
14         return code;
15     }
16
17     public void setCountry(String code) {
18         this.code = code;
19     }
20
21     public String getName() {
22         return name;
23     }
24
25     public void setName(String name) {
26         this.name = name;
27     }
28 }
```

OUTPUT:

